

Wei-Kai Li Academic CV

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Education

- 2018–2022 / PhD / Computer Science and Technology / Nanjing University of Aeronautics and Astronautics
- 2015–2018 / Master's / Computer Technology and Applications / Chongqing Jiaotong University
- 2011–2015 / Bachelor's / Information and Computing Science / Chongqing Jiaotong University

Research Projects

- Unsupervised Domain Adaptation beyond Domain Alignment and Its Extensions / NSFC Young Scientists Fund / Principal Investigator
- Advanced School in Applied Machine Learning / NSFC International Cooperation and Exchange / Principal Investigator
- Multimodal Neurophysiological Mechanisms of Heterogeneous Brain Networks in Depression and Digital-Twin Closed-Loop TMS Robotics / Shandong Major Basic Research Project / Principal Investigator
- Exploration of Unsupervised Domain Adaptation beyond Domain Alignment / MIIT Key Laboratory Open Project / Principal Investigator
- Early Identification of Parkinson's Disease Using PET/MR Multimodal Brain Networks / Hubei Key Laboratory of Molecular Imaging / Principal Investigator
- Multiscale Analysis of Autism Using Multimodal Imaging / Chongqing Education Commission Young Research Fund / Principal Investigator
- Brain Network Disease Diagnosis Based on Multicenter Transfer Learning / Chongqing Natural Science Foundation General Project / Principal Investigator
- Probabilistic Synchronization and Secure Control of Stochastic Complex Networks / Chongqing Natural Science Foundation Joint Fund / Researcher
- Investigating and Extending the Effectiveness of Self-Supervised Learning / NSFC General Program / Researcher

Publications

- 2026 / Universum Inspired Graph Contrast Learning for Major Depressive Disorder Identification: In Functional Brain Network View. H Qin, G Hong, D Ma, Z Wang, L Zhao, W Li, X Xu. CNS Neuroscience & Therapeutics 32 (9), e71158, 2026
- 2026 / Multi-Source Multi-View Graph Domain Adaptation with Hyperbolic Residual Encoding for Cross-Site MDD Identification from rs-fMRI. Z Zheng, X Chen, H Jiang, R Tian, Q Cai, J Liu, X Chen, W Li, Y Wang. arXiv preprint arXiv:2607.29531, 2026
- 2026 / Domain generalization meets mutual bi-attention: A novel approach to drug-target prediction. J Yang, W Li, K Qian, Z Cheng. Neurocomputing, 134626, 2026
- 2026 / Enhancing Fatigue Detection Through Heterogeneous Multisource Data Integration and Cross-Domain Modality Imputation. L Cui, Y Wu, Y Tang, W Li. IEEE Transactions on Computational Social Systems, 2026
- 2026 / Hierarchical neighbor integration graph attention network for autism spectrum disorder diagnosis. D Ma, L Peng, L Zhang, W Li, X Gao. Frontiers in Psychiatry 17, 1846996, 2026
- 2026 / TumorAL: Evidence-aware active learning for 3D tumor segmentation. H Wang, J Leng, S Li, Y Zhao, W Li, W Li, X Gao. Neurocomputing, 134239, 2026
- 2026 / Identification of Major Depressive Disorder Using Multiple Functional Connection Patterns. Y Ruan, L Guan, L Peng, L Zhao, Y Huang, W Li, X Gao. CNS Neuroscience & Therapeutics 32 (6), e70951, 2026

- 2026 / Electrocardiographic Alterations Combined with Hematological, Biochemical, and Metabolic Profiles Predict Prognosis in Kawasaki Disease. Q Wang, W Li, J Wan, L Wei, Y Xia, Y Hua, K Zhou, D Qie, W Li, Y Li. *Journal of Cardiovascular Development and Disease* 13 (6), 228, 2026
- 2026 / Partial domain adaptation enables cross domain cell type annotation between scRNA-seq and snRNA-seq. X Chen, Q Zou, Q Cai, X Chen, W Li, Y Wang. *PLOS Computational Biology* 22 (5), e1014223, 2026
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- 2026 / Test-time few-shot object detection via dynamic prototype fusion. Y Wu, Y Li, H Wei, W Li, Y Tang. *IEEE Transactions on Cybernetics*, 2026
- 2026 / Machine learning-based assessments of perioperative features in determining long-term conduct block post-transcatheter closure of ventricular septal defect. Q Wang, Y Xia, L Wei, P Yang, K Zhou, Y Hua, W Li, Y Li. *Cardiology in the Young*, 1-11, 2026
- 2025 / Attention Transfer Based Hybrid Knowledge Distillation for Multimodal Brain Tumor Segmentation. H Ma, HS Gan, H Mi, P Xu, Y Liu, W Li. *2025 IEEE International Conference on Big Data (BigData)*, 7503-7510, 2025
- 2025 / More insights into disruption and decoupling of individual metabolic connectomes in Parkinson's disease. D Li, J Yao, W Li, Z Guo, X Lv, L Chen, J Shi, YC Chen, J Ye. *Progress in Neuro-Psychopharmacology and Biological Psychiatry*, 111578, 2025
- 2025 / Incorporating Negative Node Information in Graph Transformer for MDD Diagnosis. H Song, H Qin, W Li. *2025 International Conference on Cyber-Physical Social Intelligence (CPSI)*, 1-6, 2025
- 2025 / MVSSL-GT: A Graph Transformer Method with Multi-View Contrastive Self-Supervised Learning. Y Lu, N Tian, W Li. *2025 International Conference on Cyber-Physical Social Intelligence (CPSI)*, 1-6, 2025
- 2025 / Low-light image enhancement using dual cross attention. Y Ruan, H Ma, D Ma, W Li, X Wang. *Engineering Applications of Artificial Intelligence* 159, 111501, 2025
- 2025 / Auto-weighted projective one-step multi-view clustering. X Mou, W Li, B Li, J Hu. *Neurocomputing*, 131960, 2025
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- 2025 / Timestamp calibration for time-series single cell rna-seq expression data. X Chen, S Lin, X Chen, W Li, Y Li. *Journal of Molecular Biology* 437 (9), 169021, 2025
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- 2024 / Synthesizing A β -PET based on multi-modal neuroimaging fusion for pathological diagnosis of Alzheimer's disease. Y Yang, B Li, W Li, H Chen, W Cao. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
- 2024 / Universum Inspired Graph Contrast Learning for Autism Spectrum Disorder Identification. H Qin, X Xu, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
- 2024 / Enhancing Functional Brain Network Construction for Neurological Disorder Diagnosis through Modularity-Driven Regularization. D Sun, X Chen, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
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- 2024 / Topology-Enhanced Multi-Modal Residual Graph Convolutional Network for Autism Spectrum Disorder Identification. J Deng, L Guan, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
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- 2024 / The characteristic patterns of individual brain susceptibility networks underlie Alzheimer's disease and white matter hyperintensity-related cognitive impairment. H Chen, J Xu, W Li, Z Hu, Z Ke, R Qin, Y Xu. Translational Psychiatry 14 (1), 177, 2024
- 2024 / TIDE: Test-time few-shot object detection. W Li, H Wei, Y Wu, J Yang, Y Ruan, Y Li, Y Tang. IEEE Transactions on Systems, Man, and Cybernetics: Systems 54 (11), 6500-6509, 2024
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- 2023 / Functional and structural brain network construction, representation and application. W Li, Z Wang, S Hu, C Chen, M Liu. *Frontiers in neuroscience* 17, 2023
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